

So then testing $\rightarrow$ 30/50% development cost

Start testing as early in development as possible

• USE CASE $\rightarrow$ sequence of actions performed by or system to provide an observable result of value to a person or another system

- Test procedures are designed for the test cases
- scripts to implement procedures.

$\rightarrow$ Devs can begin writing Test cases as soon as use cases are available / before code is written

Intro to use cases $\rightarrow$ UML

Ovals $\rightarrow$ use cases
Stick Figures $\rightarrow$ actors

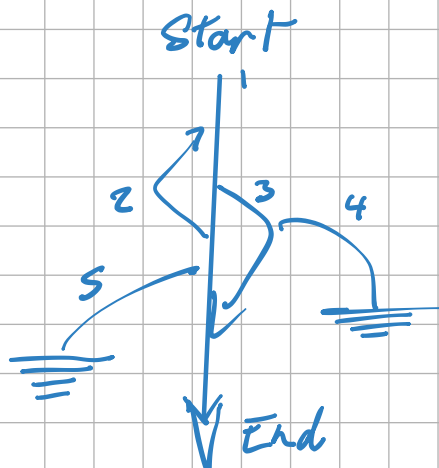


Use case diagram

Table of use case

Use Case Name	Description An appropriate name for the use case
Brief description	role & purpose
Flow of Events	textual description of flow
Special Requirement	
Preconditions	
Post conditions	

Flow of events
- basic flow
- alternate flow



Use case scenario

→ instead of a use case

→ Path

Scenario 1 Basic Flow

Scenario 2 Basic Flow / Alternate 1

⋮

Generating test cases

- test inputs

identify / communicate conditions that will be implemented in test

- Necessary to verify success

1- generate full set of use-case scenarios

2- for each scenario, identify one test case and conditions to execute

3- for each, identify data values with which to test

Step 1: Generate Scenarios

Scenario Name

Starting Flow

Alternate

Step 2: Identify Test cases

One test-case per scenario

Such as; 3A: Unfulfilled prerequisites...

Test case Matrix

Test ID	Scenario / Conditions	Student Id	Password...	Expected result
1	SC 1:	✓	✓	✓
2	SC 2:	✓	X	X
3	SC 3:	✓	✓	X
⋮	⋮			

Values → V: Valid
I: Invalid
N/A: not necessary

Step 3: Identify Data values to Test

- Review and validation

Student	password	
ID		
jheumann	abc123	→ Success
:		

Putting it all together

- Associated with the front-end of the code
- Starting early reduces costs/time